Transcript in English

Disruptive Innovation

[Visual information: The video incorporates hand-made animated illustrations over a white board as well as charts and diagrams to emphasize and help to clarify the concepts in the voiceover. The visuals are created as the voiceover introduces each new concept.]

«Brain Gear»

«Disruptive Innovation with A/Prof. Robert Perrons»

Robert Perrons:
How do disruptive technologies cause market leaders to fail, even when they seem to do everything right? Everybody knows why badly managed businesses fail, but what's more interesting is how companies like Kodak or Nortel Networks seemed to do everything right and still didn't survive.

«Prof. Clayton Christensen»

In the 1990s, Professor Clayton Christensen from the Harvard Business School put forward a theory of disruptive technology to explain this. In every market, there's a trajectory of performance improvement that customers are able to make use of. But where this trajectory lies is different from one customer to the next. Some customers can be satisfied with very basic levels of performance. Some others are very demanding and will only feel satisfied by very high levels of performance from a technology.

Disruptive technologies enter the market offering very low performance, but their performance steadily improves. So in the early days of a disruptive technology, the new innovation is considered to be not good enough by most of the market, but it seems perfectly acceptable to those customers that never really ask for more.

[Visual information: An illustration of a customer dressed casually giving a thumb up. A balloon reads, "Hey, it works for me."]

This small group of customers might be attracted by the fact that this barely good enough technology is less expensive or offers them a bit of new functionality that the old technology simply didn't provide. The problems often begin for industry incumbents when they are first assessing a potentially disruptive innovation entering their sector.

[Visual information: An illustration depicts an employee approaching his employer, who is sitting at a desk. The employee has a balloon that reads, "There's this great new thing!" The employer picks up the telephone and makes a phone call, saying, "Hey, there's this great new thing! Do you need it?" A third man is now depicted, who is standing before a professional camera on a television or movie set. The man has a balloon that says, "Meh, I've seen it and it's rubbish!"]

The company's biggest customers are telling it at that point that the new technology simply doesn't meet their needs. So, they apply the usual business logic and very rationally conclude that they should stick with the technology that they already have.

But here's the thing. The new technology that started out as low quality still gets adopted in an undemanding corner of the market, and steadily improves until it is good enough to meet the performance expectations of even the most demanding customers.

[Visual information: The man depicted previously as a less demanding customer is holding a video camera. The man previously depicted as a media professional tries to grab the video camera and has a balloon that reads, "So many great features now! I want one! I NEED one."]

By this time, however, it's usually too late for the incumbent to do much about it because they failed to develop the required capabilities and linkages to leverage the new technology.

[Visual information: The employer on the previous picture is on the phone, showing an angry face and asking, "What on earth is going on?" The media professional is now depicted with a frustrated look and a balloon that says, "Have you guys been hiding under a rock?"]

Christensen frequently uses the advent of the word "processor" as an example. Many people who used them in the 1980s probably remember being frustrated by the fact that the computer sometimes struggled to keep up with your fingers as you were typing.

Demanding customers like professional secretaries would have been horrified by the new technology and the typewriter company would have rationally listened to these customers and decided that they should continue to stay focused on improving typewriters.

[Visual information: An illustration of a woman with a balloon that says, "But I can delete a wrong letter or word with one key-stroke!" A second woman is standing by her side and says, "That's because you're not good enough. This will never work for professionals!"]

«Fast-forward fifteen years or so!»

Despite their imperfect performance, however, these early word processors had tremendous appeal for one group of customers, people who didn't type very well and who loved being able to easily fix their mistakes. This group was willing to forgive the lack of speed in exchange for this new functionality.

But as computers got better and faster, word processors steadily improved until even the fastest, most demanding typists were satisfied with the new technology. Now think about technologies that are unfolding their disruptive potential right now, like 3D printing.

[Visual information: A man is using a large 3D printer to produce a chess piece. A woman picks up the finished piece and says, "It feels a bit… cheap."]

At the moment, 3D printers for home use produce relatively low-resolution, fused, deposition-modelled plastic compound artefacts. But give it a few years and think about the number of objects you would love to print.

[Visual information: A couple is sitting at a dinner table, with the dishes served with chicken and vegetables and a lit candle glowing in the middle of the table.]

Lost a rook in a chess game? Forgot a present, it's 10pm and the shops are closed? When affordable 3D printing can service more sophisticated ends of the market, retail, manufacturing and logistics won't be the same anymore.

One of the interesting things about disruptive innovations is that they tend to have almost nothing to do with how radical the new technology is. Instead, what makes a technology disruptive is the fact that it isn't sustaining, which basically means that it is not based on the same markets and value networks as what is already available.

[Visual information: A picture depicts the media professional and the amateur cameraman having a conversation. The amateur cameraman says, "The tech in this new model is amazing, and so disruptive!" The media professional says, "It's not the tech that's disruptive Jimmy, it's that boss man up there doesn't get it!"]

In that regard, the emergence of disruptive innovations is really almost as much of a management phenomenon as it is about the technology in and of itself. Disruptive innovations get a lot of attention in the business press because of the seismic shifts that they can deliver in industries. In this way, disruptive innovations are sometimes a force for renewal in the marketplace. And while they are responsible for failures of large companies, every one of these fatalities in the business world tends to be a new entrepreneur's success story.

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